

Functional Requirements Document (FRD)

1. Project Title

Apple Retail Sales & Warranty Analytics Dashboard

2. Dashboard Sections

The dashboard will consist of five main pages: 1. Overview Dashboard 2. Sales Performance Dashboard 3. Product Analysis Dashboard 4. Store Performance Dashboard 5. Warranty & Claims Dashboard

Each page will display KPIs, charts, and filters relevant to the analysis theme.

3. Data Requirements

Sales Table

- sale_id
- sale_date
- store_id
- product_id
- quantity
- price
- total
- year, month

Products & Categories

- product_id
- product_name
- category_id
- category_name
- launch_date
- price

Stores

- store_id
- store_name
- city
- country
- region

Warranty Claims

- claim_id
- claim_date
- sale_id
- repair_status
- year, month

4. Filters / Slicers

- Year
- Month
- Region
- Country
- Category
- Product
- Repair Status

5. Visuals / Charts

Overview Dashboard

- KPI Cards: Revenue, Quantity, Orders, Stores, Claim Rate
- Bar Chart: Revenue by Category
- Pie Chart: Revenue by Region
- Line Chart: Year-wise Revenue Trend

Sales Performance

- KPI Cards: Revenue, Quantity, Orders, AOV
- Map: Revenue by Country
- Column Chart: Monthly Sales
- Bar Chart: Top Cities by Revenue
- Gauge: Target vs Actual Revenue

Product Analysis

- KPI Cards: Total Products, Total Quantity Sold
- Bar Chart: Top 10 Products
- Bar Chart: Lowest Selling Products
- Table: Product Summary with Sales & Claims

Store Performance

- KPI Cards: Sales, Quantity, Orders, AOV
- Bar Chart: Top 10 Stores
- Table: Store Summary
- Column Chart: Stores by Country

Warranty & Claims

- KPI Cards: Claim Count, Claim Rate
- Bar Chart: Claims by Product
- Pie Chart: Claimed vs Not Claimed
- Bar Chart: Claims by Category
- Cards: Repair Status Counts

6. Interactivity

- Drill-down: Category → Product
- Cross-filtering between visuals
- Tooltips with additional metrics
- Bookmarks for:
 - Filter View / Clean View
- KPI Toggle (Revenue / Quantity)

7. Calculations / Measures

- Total Revenue = $\text{SUM}(\text{Sales}[\text{Total}])$
- Total Quantity = $\text{SUM}(\text{Sales}[\text{Quantity}])$
- Total Orders = $\text{DISTINCTCOUNT}(\text{Sales}[\text{sale_id}])$
- Average Order Value = $\text{Revenue} / \text{Orders}$
- Warranty Claim Rate = $\text{Claims} / \text{Orders}$
- Highest & Lowest Product Revenue
- Year-over-Year Growth %

8. Export / Reporting Needs

- Power BI Service Dashboard Sharing
- PDF export for management review
- Optional Excel export for tables

9. Notes / Special Instructions

- Follow Apple-style clean UI (white theme)
- Maintain consistent color coding across dashboards
- Ensure performance optimization using proper relationships and measures